

MUHAMMAD ZAIN AKRAM

Full-Stack Software Engineer | AI Integration (RAG, LLMs, ML)

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 5 years of experience building modern web applications using c+, JavaScript, React.js, Next.js, Node.js, Python, and Django REST Framework. Experienced in developing responsive user interfaces, REST APIs, authentication, database-driven applications, and AI-powered features including RAG, LLMs, and OCR. Focused on clean, maintainable code, performance, scalable architecture, and reliable applications from development through production.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Three.js, Framer Motion, Bootstrap

Backend: Node.js, Express.js, Django REST Framework, REST API Design

Databases: PostgreSQL, MongoDB, SQLite

AI / Machine Learning: Google Gemini API, LangChain, RAG (Retrieval-Augmented Generation), ChromaDB, TensorFlow/Keras (LSTM, RNN), Scikit-learn (Random Forest, Gradient Boosting, Linear Regression), OpenCV, Tesseract OCR, Pandas, NumPy, Prompt Engineering

Tools & DevOps: Git, GitHub, Docker, Docker Compose, JWT Authentication, Vercel, Agile/Scrum

PROFESSIONAL EXPERIENCE

Full-Stack Developer Intern — Terasols

2025

Next.js · .NET Core 8 · Python (FastAPI) · PostgreSQL · RabbitMQ · Redis/Celery · Docker

- Spearheaded the development of an Enterprise Task Management System; implemented role-based dashboards and JWT authentication, improving system security and user access by 30%.
- Built the analytics and reporting microservice in FastAPI (performance trends, workload distribution), integrated alongside a .NET Core API handling authentication and task CRUD.
- Containerized the multi-service architecture (Next.js frontend, two backend services, PostgreSQL, RabbitMQ, Redis) with Docker Compose for local and cloud deployment.

PROJECTS

AI-Based Smart Energy Consumption Prediction System — Final Year Project

Mar 2025 – Aug 2025

Django REST Framework · React (Vite) · Three.js · Scikit-learn · TensorFlow/Keras · OpenCV · Tesseract OCR · LangChain · ChromaDB · Google Gemini API

- Built an end-to-end AI web application forecasting next-month household energy consumption and billing from historical usage data, with personalized energy-saving recommendations.

- Trained and benchmarked five forecasting models — Linear Regression, Random Forest, and Gradient Boosting (Scikit-learn) alongside LSTM and RNN networks (TensorFlow/Keras).
- Built an OCR pipeline (OpenCV + Tesseract) that automatically extracts consumption readings from uploaded utility bill images, removing manual data entry.
- Developed SECPARS, a Retrieval-Augmented Generation chatbot using LangChain and ChromaDB (sentence-transformer embeddings) with the Google Gemini API, enabling natural-language Q&A on usage and predictions.

PakBooking — Hotel Booking Platform
Sep 2025

Django REST Framework · Next.js · PostgreSQL · Docker · JWT · Stripe/Razorpay

- Architected a production-style full-stack hotel booking platform for Pakistan's hospitality market with JWT authentication and role-based access control for users and admins.
- Built the complete booking lifecycle (Pending → Confirmed → Completed → Refunded) with automated notifications and a tiered refund policy.
- Integrated Stripe/Razorpay payment flows and an admin analytics dashboard; reached a 90+ Lighthouse score through PostgreSQL query optimization and PWA offline support.
- Shipped with Docker Compose, Swagger/OpenAPI documentation, and English/Urdu/Roman Urdu localization.

Creative 3D Developer Portfolio — Personal Project
2026

Next.js 16 · React 19 · Three.js (@react-three/fiber, drei) · Tailwind CSS 4 · Framer Motion

- Designed and shipped an interactive 3D portfolio with custom 3D models, scroll-driven animation, and a working contact form (React Hook Form + Nodemailer).
- Resolved production issues including a Server/Client Component rendering crash, and self-hosted forked GitHub-Readme-Stats and Streak-Stats instances to bypass shared-API rate limits on live GitHub stat cards.
- Deployed on Vercel using Bun as the package manager.

Online Banking System — University Project
Jun 2025

MongoDB · Express.js · React.js · Node.js (MERN Stack)

- Built a secure MERN-stack banking application supporting account creation, fund transfers, bill payments, and real-time transaction tracking.
- Implemented RESTful APIs with JWT authentication and role-based dashboards for users and admins.

EDUCATION

B.S. Computer Science — Lahore Garrison University, Lahore, Pakistan
2021 – 2025

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Artificial Intelligence, Machine Learning, Deep Learning, Web Development & Design, Computer Networks

Intermediate (Pre-Engineering) — Al-Zahrawi Science College, Lahore
2018 – 2020

CERTIFICATIONS

- **Claude 101** — Anthropic (2025): fundamentals of working with Claude AI, prompt design, and responsible AI usage.
- **AI Fluency: Framework & Foundations** — Anthropic, in partnership with University College Cork, Ringling College of Art + Design, HEA & National Forum (2025).

ADDITIONAL INFORMATION

Languages: English (Professional), Urdu (Native)

Open Source: Active GitHub contributor (github.com/zainakramwork4) — self-hosted and forked developer-tooling projects (GitHub Readme Stats, Streak Stats) to production.